TBA3241 Social Media Analytics

AY25/26 Semester 2

Final Report

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Executive Summary

<to be added after analysis is done>

Data & Network Construction

Data Source

This study uses the Twitter Interaction Network for the U.S. Congress, obtained from the Stanford Network Analysis Project (SNAP). The dataset consists of publicly available Twitter interactions among members of the 117th U.S. Congress.

The data was originally collected using Twitter's API, which the researchers used the Tweepy Python package to interact with, retrieving the most recent 3,200 tweets (the limit set by the API) as of data collection date on 9 June 2022 (Fink, Omodt, Zinnecker, & Sprint, 2023).

Data Cleaning

From the above 3,200 tweets, some data cleaning was done by the researchers. Tweets that were created outside of a common time window (9 February 2022 to 9 June 2022) were excluded. Tweets made by Congressional members who tweeted less than 100 times during the same time window were also excluded (Fink, Omodt, Zinnecker, & Sprint, 2023). The final dataset contains 475 nodes and 13,289 edges after analysing the interactions amongst the remaining tweets.

Due to the above data cleaning, there are two types of sampling bias present in the dataset. The first type of bias is activity bias, where highly active users are over-represented since tweets made by less active users were excluded. The second type of bias is temporal sampling bias, where the dataset only captures a fixed time window and not long term behaviour.

Network Construction

The network was constructed through the following pipeline. First, tweets and its data were extracted using Twitter API. After data cleaning, interactions between use2rs were identified. These interactions include retweets, replies, mentions, and quote tweets. They are then aggregated to obtain the total number of interactions between each pair of users. These interaction counts were normalised by the total number of tweets produced by source user to derive influence probabilities.

The network is a directed, weighted, and unimodal network, stored as a weighted directed edge list. The nodes represent individual Congress member's Twitter account, and the directed edges represent the interaction initiated by target node to source node, where interaction refers to be retweets, quote tweets, replies, and mentions. The edge weights represent the probability of influence. This is

calculated using the ratio of number of interactions to the total number of tweets issued by source node.

Datasets

Congressional Twitter network

The Twitter interaction network for the 117th United States Congress House of Representatives was constructed by first obtaining members' official Twitter handles from `https://pressgallery.house.gov/member-data/members-official-twitter-handles`. The Twitter API was then used to obtain all Tweets by members of Congress between February 9, 2022, and June 9, 2022. (The latter date corresponds to when the data were requested, and the earlier date was the farthest point backward in time at which one user's number of Tweets exceeded the 3200-Tweet limit set by the API.) Only the 475 representatives (out of 535) who issued 100 or more tweets during this time frame were included in the network.

A weighted, directed network was constructed such that the connection from user j to user i was given by

$$\mathcal{P}_{ij} = \frac{n_{ij}^{\text{retweet}} + n_{ij}^{\text{quote}} + n_{ij}^{\text{reply}} + n_{ij}^{\text{mention}}}{N_j^{\text{tweets}}}, \qquad (7)$$

where n_{ij}^{retweet} is the number of times user i retweeted user j, n_{ij}^{quote} is the number of times user i quote tweeted user j, n_{ij}^{reply} is the number of times user i replied to user j, n_{ij}^{mention} is the number of times user i mentioned user j, and N_j^{tweets} is the total number of Tweets issued by user j. The weighted connection, $\mathcal{P}_{ij}$, can therefore be interpreted as the probability that user i is observably influenced by user j.

Network Analysis

<with necessary visualisation presented>

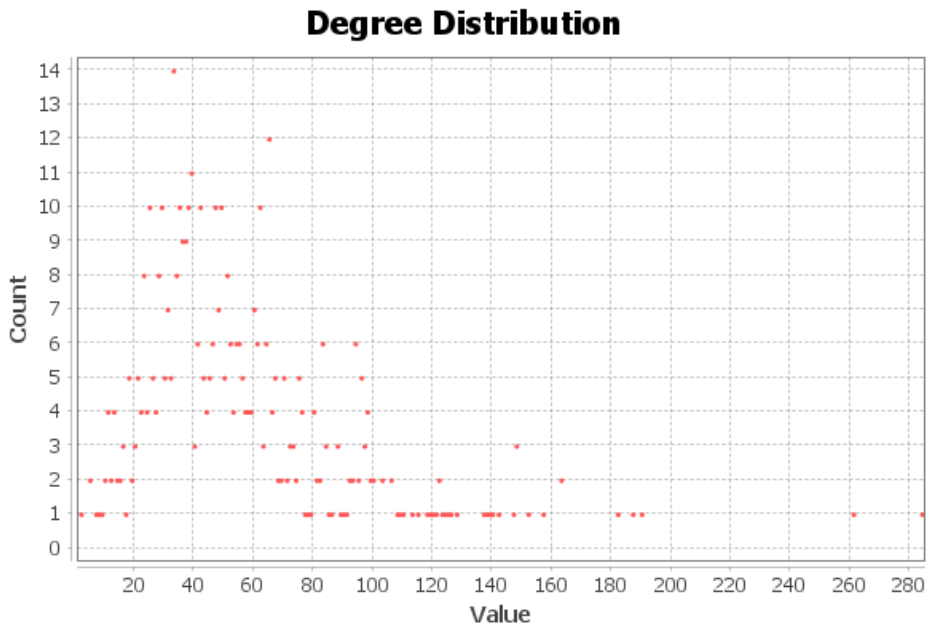
Network Structure

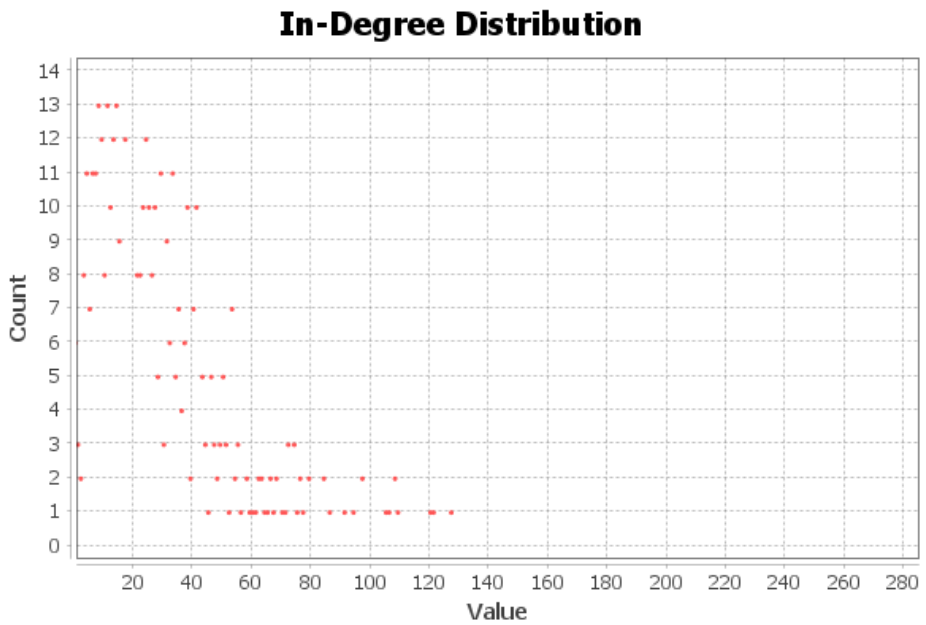
<descriptive network structure, e.g. size, density, diameter>

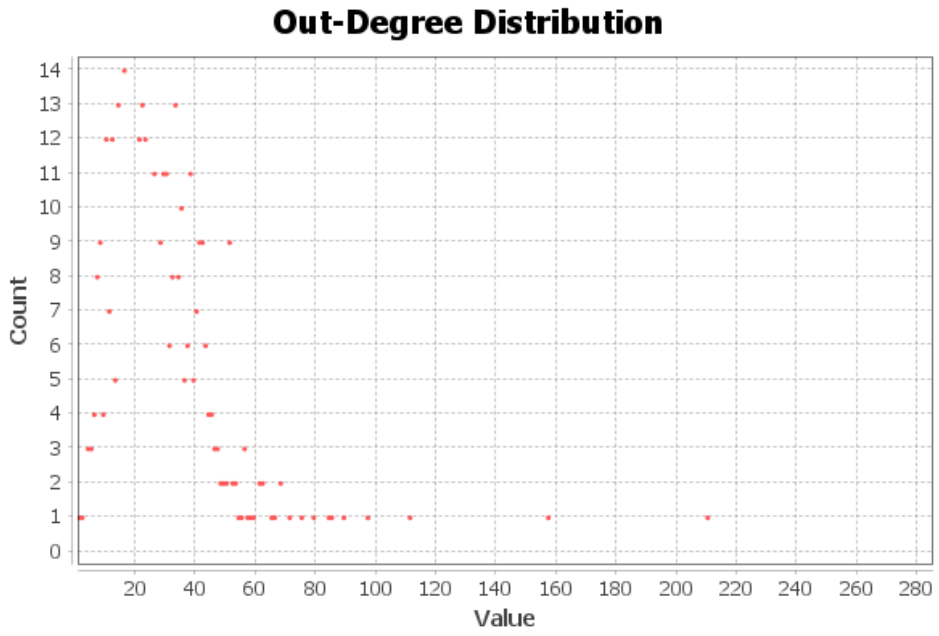
<node level attributes, e.g. centrality measures>

<community structure: community (e.g. cluster) detection and interpretation>

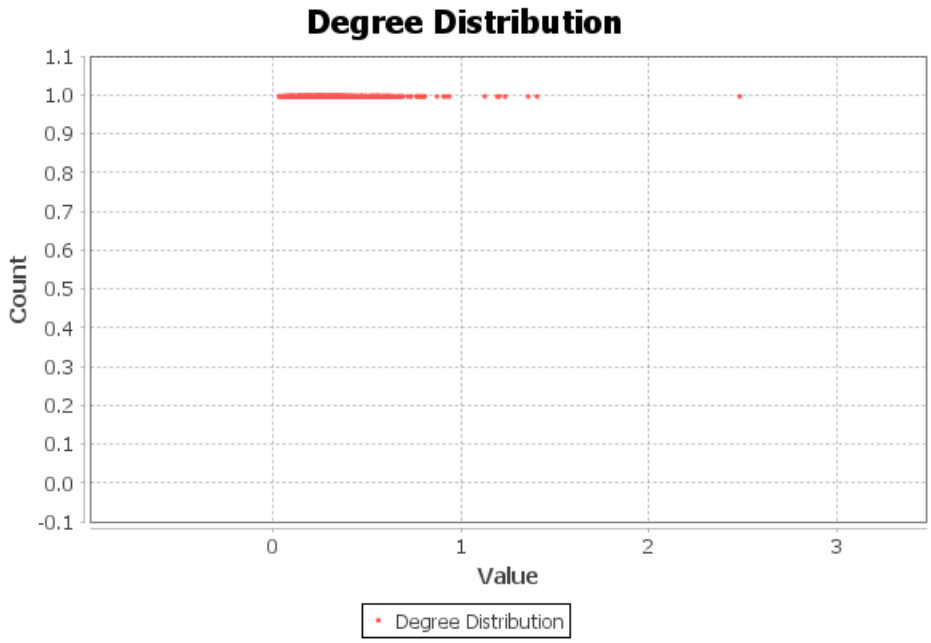
Average Degree: 27.977

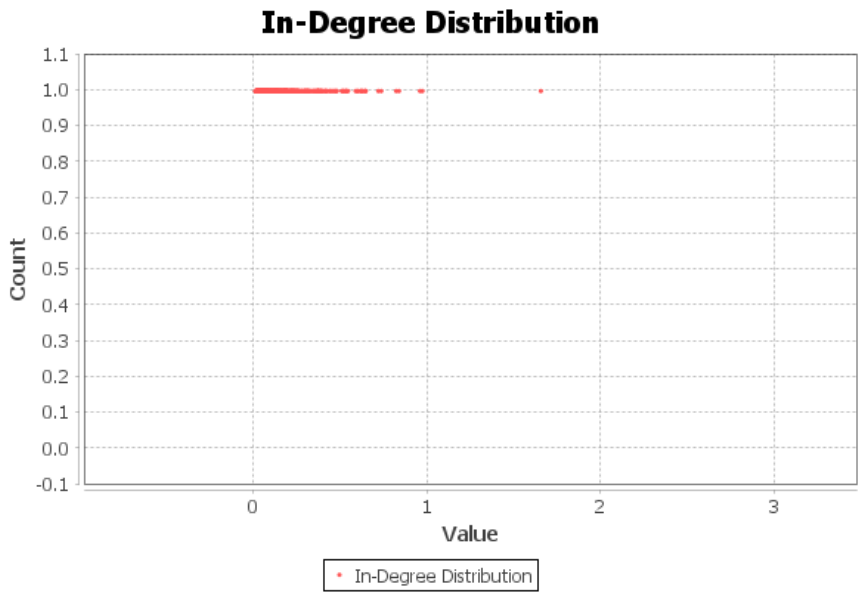


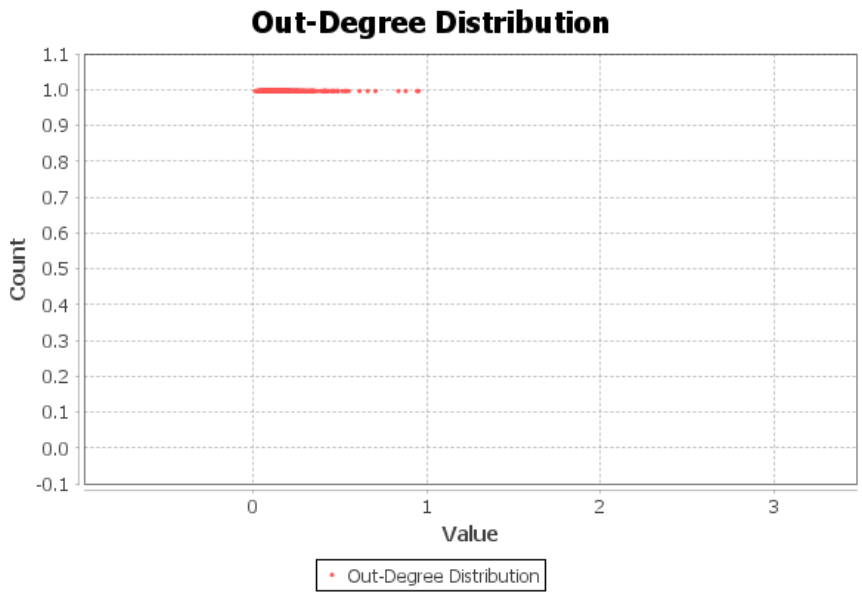




Average Weighted Degree: 0.163



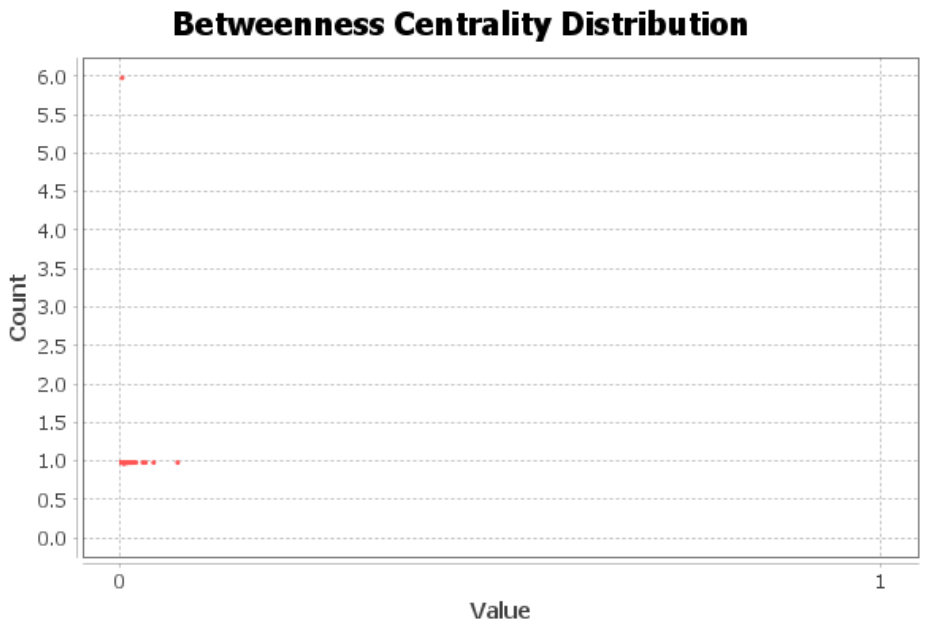


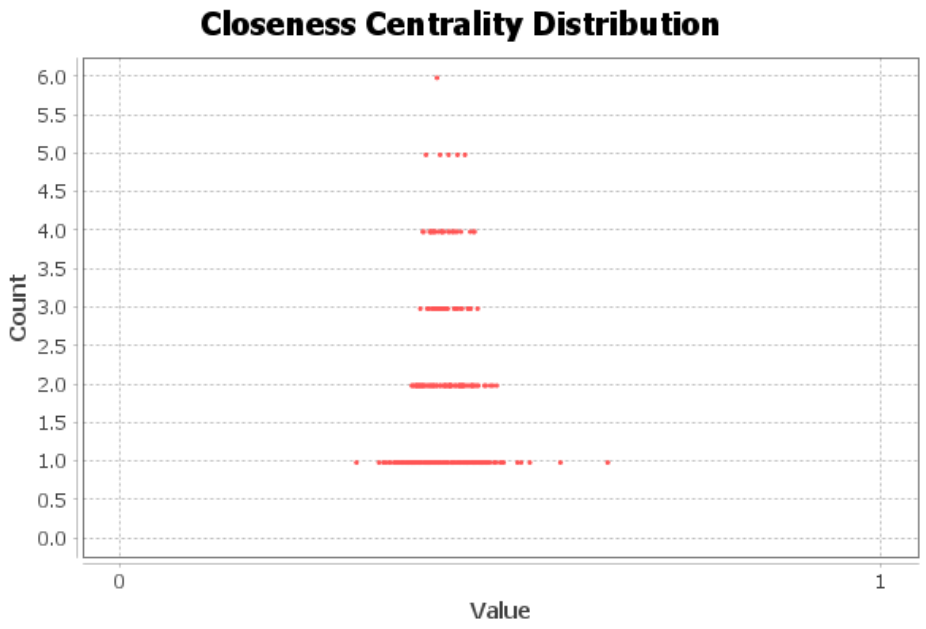


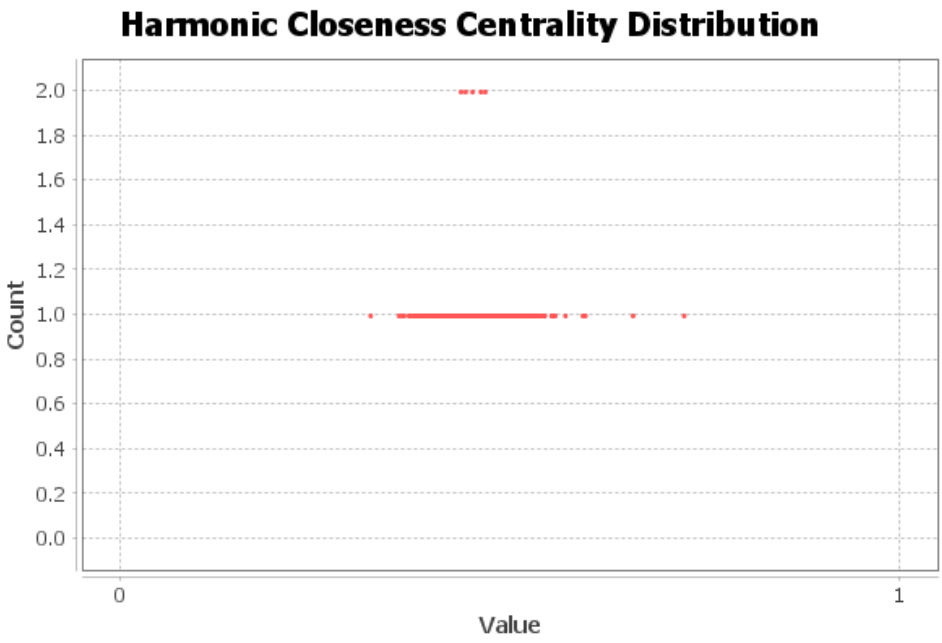
Diameter: 6

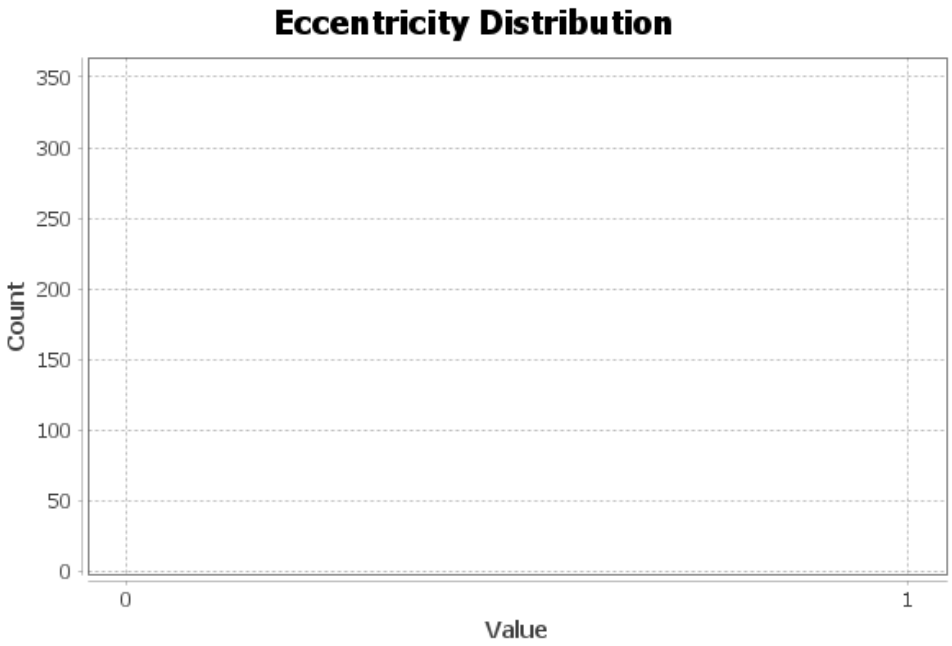
Radius: 3

Average Path length: 2.357151853751136


Betweenness Centrality Distribution
Count
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
Value


Closeness Centrality Distribution
Count
6.0
5.5
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
0
1
Value





Density: 0.059

Number of Weakly Connected Components: 1

Number of Strongly Connected Components: 7

Average Clustering Coefficient: 0.209 (The Average Clustering Coefficient is the mean value of individual coefficients)

<choose one egocentric network to examine and analyse its structure and dynamics>

Research Questions

Community Differences

<research qn: community differences

- How can the nodes be categorised?
- How do the categories differ from one another?
- Is the Congress Twitter network structurally centralised around a few dominant members?
- Does the network exhibit high degree centralisation or betweenness centralisation?
- Can compare democratic vs republic, senate vs house>

Focus on Influence

<research qn: focus on influence (By identifying the "influential" users in this network, you can also further analyse whether the network level influence is aligned with their real role status in the congress in reality. There might be some interesting findings.)

- Who are the most influential users, and why?
- Is there an imbalance of influence where influence is not reciprocated?
- Are the influential users balanced across parties?
- Can use statistics we learnt in class to examine the centrality and network structure>

Cross Party Influence

<research qn: cross party influence

- To what extent are there cross party influence?
- Are there any critical bridging nodes that act as articulation point?
- Is there a stronger influence within the same party compared to cross party?
- Can potentially evaluate within-group influence vs cross party influence>

What-if Questions

Removal of nodes

How will the overall network cohesion (e.g. number of components, diameter, fragmentation) change if the most influential nodes are removed?

To examine whether the congressional Twitter network heavily relies on a small number of influential users, we conducted a what-if analysis by removing the top-ranked nodes based on viral centrality. Viral centrality was chosen because it reflects a node's ability to spread influence within a weighted directed network, where edge weights represent transmission probabilities. We compared the network structure after removing the top 1, 5, and 10 influential nodes with the baseline structure.

In the baseline network, the graph consists of 475 nodes and 13,289 directed edges, forming a single weakly connected component. The largest weakly connected component includes all nodes, with fragmentation equal to 0 and a weak projection diameter of 4, indicating a cohesive structure.

```
--- BASELINE NETWORK ---
Nodes: 475
Edges: 13289
Weak components: 1
Largest weak component: 475
Fragmentation: 0.0000
Diameter (weak projection): 4
```

```
Top 10 influential nodes (Viral Centrality):
 node_id         Users      Full Name      Party Chamber State/District  viral_centrality
     399  SteveScalise  Steve Scalise Republican   House           LA01          1.145936
     367 SpeakerPelosi   Nancy Pelosi   Democrat   House           CA12          1.112454
     393  RepBobbyRush     Bobby Rush   Democrat   House           IL01          1.007327
     322     GOPLeader Kevin McCarthy Republican   House           CA23          0.979915
     470  RepJoeWilson     Joe Wilson Republican   House           SC02          0.810477
     164 RepJamesComer    James Comer Republican   House           KY01          0.784266
     436 RepMarkTakano    Mark Takano   Democrat   House           CA41          0.780674
     226    RepBobGood       Bob Good Republican   House           VA05          0.687173
     159    RepCloudTX  Michael Cloud Republican   House           TX27          0.674471
     208   RepFranklin Scott Franklin Republican   House           FL15          0.646444

Party composition among top 10 influential nodes:
Party
Republican    7
Democrat      3

Chamber composition among top 10 influential nodes:
Chamber
House    10
```

Across all removal scenarios, the network remains as a single weakly connected component, with fragmentation unchanged at 0 and diameter remaining constant at 4. Although the number of edges decreases, the reduction in component size is minimal, and the overall structure remains stable.

```
==============================
WHAT-IF 1A: REMOVE INFLUENTIAL NODES
==============================

--- After removing top 1 influential nodes ---
Nodes: 474
Edges: 13137
Weak components: 1
Largest weak component: 474
Fragmentation: 0.0000
Diameter (weak projection): 4

--- After removing top 5 influential nodes ---
Nodes: 470
Edges: 12349
Weak components: 1
Largest weak component: 470
Fragmentation: 0.0000
Diameter (weak projection): 4

--- After removing top 10 influential nodes ---
Nodes: 465
Edges: 11951
Weak components: 1
Largest weak component: 465
Fragmentation: 0.0000
Diameter (weak projection): 4
```

These findings indicate that removing even the most influential nodes does not disrupt the global connectivity of the network. While these nodes contribute significantly to information diffusion, they are not structurally critical for maintaining network cohesion.

Removal of nodes

What if the bridging nodes are removed? Does the network rely on a small subset of nodes to maintain connectivity?

To assess whether the network relies on a few bridging nodes for connectivity, we conducted a what-if analysis by removing the node with the highest betweenness centrality. Betweenness centrality identifies nodes located on multiple shortest paths, which are potential bridges connecting different parts of the network.

The top 10 bridging nodes are Kevin McCarthy, Joe Wilson, Steve Scalise, Sean Casten, Nancy Pelosi, Rick Scott, Chip Roy, John Katko, Bobby Rush and August Pfluger. In terms of political affiliation, 7 of the top 10 are Republicans and 3 are Democrats. In terms of institutional role, 9 of the top 10 are members of the House of Representatives, and only one is a senator. This indicates that the most influential bridging nodes in the network are concentrated among members of the House of Representatives, especially some well-known prominent Republican accounts.

```
Computing bridging nodes (betweenness on weak projection)...

Top 10 bridging nodes (Betweenness):
 node_id         Users      Full Name      Party Chamber State/District  betweenness
     322     GOPLeader Kevin McCarthy Republican   House           CA23     0.169548
     470  RepJoeWilson     Joe Wilson Republican   House           SC02     0.078057
     399  SteveScalise  Steve Scalise Republican   House           LA01     0.075934
     147     RepCasten    Sean Casten   Democrat   House           IL06     0.057515
     367 SpeakerPelosi   Nancy Pelosi   Democrat   House           CA12     0.057038
      72  SenRickScott     Rick Scott Republican  Senate        Florida     0.056717
     389    RepChipRoy       Chip Roy Republican   House           TX21     0.046202
     274  RepJohnKatko     John Katko Republican   House           NY24     0.046041
     393  RepBobbyRush     Bobby Rush   Democrat   House           IL01     0.045972
     370    RepPfluger August Pfluger Republican   House           TX11     0.044957

Party composition among top 10 bridging nodes:
Party
Republican    7
Democrat      3

Chamber composition among top 10 bridging nodes:
Chamber
House     9
Senate    1

Articulation points in weak projection:
None
```

The analysis shows that even after removing the top 1, 5, 10 and 24 bridging nodes, the network continues to form a single weakly connected component. Fragmentation remains at 0 and the diameter stays unchanged at 4 across all scenarios. Although the number of edges decreases and the largest component size reduces slightly, no structural fragmentation is observed.

```
--- After removing top 1 bridging nodes ---
Nodes: 474
Edges: 13005
Weak components: 1
Largest weak component: 474
Fragmentation: 0.0000
Diameter (weak projection): 4

--- After removing top 5 bridging nodes ---
Nodes: 470
Edges: 12389
Weak components: 1
Largest weak component: 470
Fragmentation: 0.0000
Diameter (weak projection): 4

--- After removing top 10 bridging nodes ---
Nodes: 465
Edges: 11844
Weak components: 1
Largest weak component: 465
Fragmentation: 0.0000
Diameter (weak projection): 4

--- After removing top 24 bridging nodes ---
Nodes: 451
Edges: 10655
Weak components: 1
Largest weak component: 451
Fragmentation: 0.0000
Diameter (weak projection): 4
```

This finding is further supported by the articulation point analysis on the weak projection, which identified no articulation points. This means there was no single node whose removal would disconnect the network. In other words, although some nodes occupy relatively stronger bridging positions than others, none of them functions as a critical cut vertex for the network as a whole.

```
Articulation points in weak projection:
None
```

Overall, the findings indicate that the network does not rely on a small subset of bridging nodes to maintain connectivity. Instead, the structure is supported by multiple redundant connections, allowing it to remain intact even when key bridging nodes are removed.

Changes to edge weight

What if the transmission probabilities are increased or decreased in different scenarios? How will this change the graph analysis and the viral centrality of the nodes?

To investigate how changes in transmission strength affect influence within the network, we conducted a what-if analysis by adjusting the edge weights representing transmission probabilities. We applied a scaling factor to simulate different levels of interaction intensity and recalculated the viral centrality for each case.

The results show that viral centrality increases as transmission strength increases. When the scaling factor rises from 0.5 to 1.5, both the average and maximum viral centrality increase significantly, indicating that stronger transmission probabilities enhance the spread of influence. Conversely, lower transmission probabilities reduce the overall diffusion potential.

Despite these changes in magnitude, the ranking of influential nodes remains largely stable across all scenarios. Key users such as Steve Scalise, Nancy Pelosi and Bobby Rush consistently appear among the top-ranked nodes. This suggests that increasing transmission strength amplifies influence but does not significantly alter the overall hierarchy of influence.

Importantly, adjusting the transmission probabilities does not change the network structure because the nodes and connections remain unchanged. Therefore, structural properties such as connectivity are unaffected; the impact is primarily on the strength of information diffusion. In summary, these findings suggest that while transmission strength affects how widely information spreads, the underlying network structure plays a more fundamental role in determining which nodes are most influential.

```
Beta = 0.5
Mean Viral Centrality: 0.0886
Max Viral Centrality: 0.5143
Top node: SteveScalise | Republican | House | VC=0.5143
Top 5 nodes:
 node_id         Users      Party Chamber  viral_centrality
     399  SteveScalise Republican   House          0.514345
     367 SpeakerPelosi   Democrat   House          0.507917
     393  RepBobbyRush   Democrat   House          0.464922
     322     GOPLeader Republican   House          0.445977
     470  RepJoeWilson Republican   House          0.372310
```

```
Beta = 0.8
Mean Viral Centrality: 0.1499
Max Viral Centrality: 0.8743
Top node: SteveScalise | Republican | House | VC=0.8743
Top 5 nodes:
 node_id         Users      Party Chamber  viral_centrality
     399  SteveScalise Republican   House          0.874328
     367 SpeakerPelosi   Democrat   House          0.859892
     393  RepBobbyRush   Democrat   House          0.781705
     322     GOPLeader Republican   House          0.756614
     470  RepJoeWilson Republican   House          0.627922
```

```
Beta = 1.0
Mean Viral Centrality: 0.1952
Max Viral Centrality: 1.1459
Top node: SteveScalise | Republican | House | VC=1.1459
Top 5 nodes:
 node_id         Users      Party Chamber  viral_centrality
     399  SteveScalise Republican   House          1.145936
     367 SpeakerPelosi   Democrat   House          1.112454
     393  RepBobbyRush   Democrat   House          1.007327
     322     GOPLeader Republican   House          0.979915
     470  RepJoeWilson Republican   House          0.810477
```

```
Beta = 1.2
Mean Viral Centrality: 0.2446
Max Viral Centrality: 1.4394
Top node: SteveScalise | Republican | House | VC=1.4394
Top 5 nodes:
     470  RepJoeWilson Republican   House          1.004747
```

```
Beta = 1.5
Mean Viral Centrality: 0.3277
Max Viral Centrality: 1.9421
Top node: SteveScalise | Republican | House | VC=1.9421
Top 5 nodes:
 node_id         Users      Party Chamber  viral_centrality
     399  SteveScalise Republican   House          1.942127
     367 SpeakerPelosi   Democrat   House          1.843910
     393  RepBobbyRush   Democrat   House          1.646070
     322     GOPLeader Republican   House          1.630453
     164 RepJamesComer Republican   House          1.370302
```

Implications

<what do the findings imply in the real world context>

(from proposal: We can potentially extend our analysis to better understand the larger influence structure of the Congress in reality, and apply the same questions in political analyses, such as finding out who exerts most influence, the cluster strength, and structural fragility. Political parties could also

potentially use the findings to strengthen intra- and inter-party influence (e.g., who is working as the spokesman on the network for each party?), as well as optimising strategies during election campaigns.)

<identify at least 2 actionable insights, e.g. platform policy, marketing, platform governance, community management>

Conclusion and Limitation

<summarise key findings>

This network is highly robust to targeted removal. Removing the most influential node reduces the number of edges but does not fragment the network. The network remains a single weakly connected component, with fragmentation remaining at zero and diameter unchanged, indicating a robust and redundant structural backbone.

The network shows strong structural robustness under multiple what-if scenarios. Removing influential or bridging nodes reduces the number of edges but does not fragment the network, which remains as a single weakly connected component with a stable diameter and no articulation points. This suggests that the network does not rely on a small set of critical nodes and is supported by redundant connections. Changes in transmission probabilities affect the strength of information diffusion, with higher values increasing viral centrality and lower values reducing it, while the overall ranking of influential nodes remains largely unchanged. This implies that influence is driven more by network structure than interaction strength.

<discuss the limitations in this work>

Reproducibility

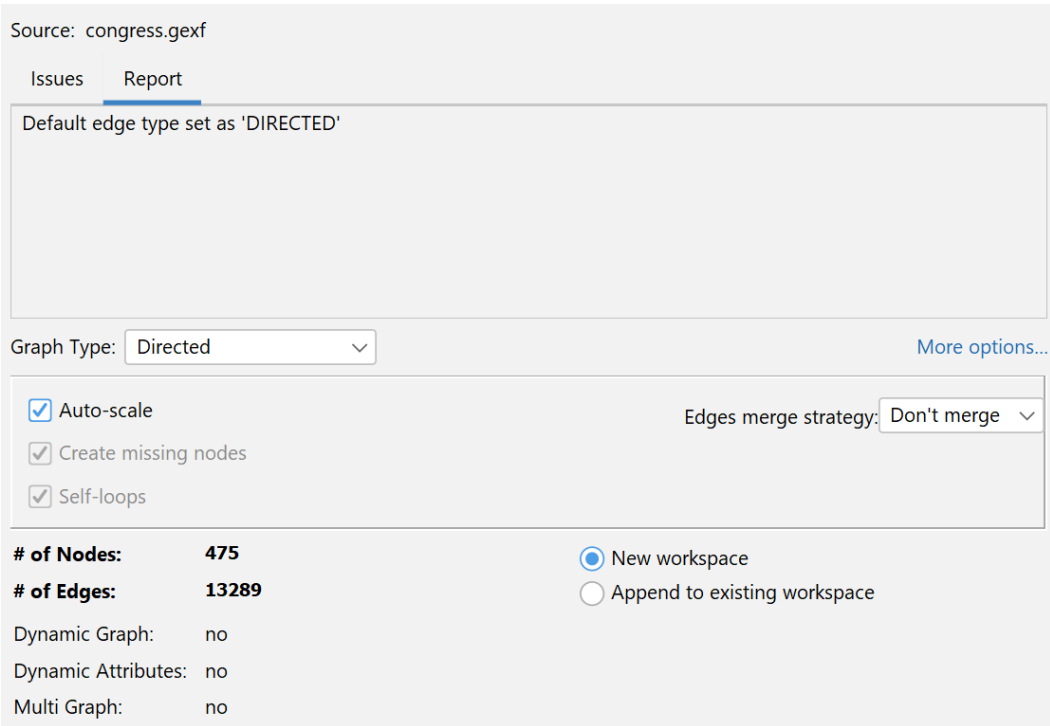
The files provided by SNAP did not contain any Gephi compatible file. To open the network in Gephi, we have to convert it to GEXF file. This can be done by converting edgelist file using Python's NetworkX library.

```python
import networkx as nx

G = nx.read_edgelist(
    'congress.edgelist',
    create_using=nx.DiGraph(),
    nodetype=int,
    data=True
)

for u, v, data in G.edges(data=True):
    data['weight'] = float(data.get('weight', 1.0))

nx.write_gexf(G, 'congress.gexf')

print("Nodes:", G.number_of_nodes())
print("Edges:", G.number_of_edges())
```

The script reads from edgelist file, constructs it as a directed graph, and writes the graph into a GEXF file.

```
<edges>
  <edge source="0" target="4" id="0" weight="0.002105263157894737" />
  <edge source="0" target="12" id="1" weight="0.002105263157894737" />
  <edge source="0" target="18" id="2" weight="0.002105263157894737" />
  <edge source="0" target="25" id="3" weight="0.004210526315789474" />
  <edge source="0" target="30" id="4" weight="0.002105263157894737" />
```

The resulting GEXF file retains edge information like direction and weight. We can then open this graph in Gephi for layout and visual analysis.



Reproducing what-if analysis

The what-if analysis was implemented using Python, primarily with the NetworkX, pandas and NumPy libraries. The main script ‘what_if_analysis.py’ loads the directed network from the ‘congress.edgelist’ file to construct a weighted graph, while additional metadata from ‘users.xlsx (Sheet2)’ is merged to incorporate attributes such as party, chamber and state for more detailed analysis. Viral Centrality is computed using the provided ‘viral_centrality.py’ algorithm, which takes input from the ‘congress_network_data.json’ file to simulate diffusion-based influence.

The analysis then evaluates multiple scenarios, including the removal of influential nodes, removal of bridging nodes, random node removal as a benchmark, and changes in transmission probabilities through a scaling factor. For each scenario, key network metrics such as number of components, largest weakly connected component, fragmentation and diameter are recalculated and displayed in the terminal. The script can be executed directly to reproduce all reported results.

References

Fink, C. G., Omodt, N., Zinnecker, S., & Sprint, G. (2023). A Congressional Twitter network dataset quantifying pairwise probability of influence. *Elsevier*.

Fink, C., Fullin, K., Gutierrez, G., Omodt, N., Zinnecker, S., Sprint, G., & McCulloch, S. (2023). A centrality measure for quantifying spread on weighted, directed networks. *Physica A*.

Stanford University. (n.d.). *SNAP: Network datasets: Twitter Interaction Network for the US Congress*. Retrieved from SNAP: https://snap.stanford.edu/data/congress-twitter.html

Declare of AI tool usage

AI Tool used	**How is the tool used in the work**
ChatGPT	This tool helps to understand concepts such as transmission probability and viral centrality, and provides general guidance for implementing network analysis techniques in Python. It was also used to refine grammar, sentence structure and clarity of explanations in the report, without altering the underlying analysis or results. All analysis, modelling, and interpretation were conducted independently by the group.